



# Cambridge International AS & A Level

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**MATHEMATICS****9709/32**

Paper 3 Pure Mathematics 3

**October/November 2024****1 hour 50 minutes**

You must answer on the question paper.

You will need: List of formulae (MF19)

**INSTRUCTIONS**

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- If additional space is needed, you should use the lined page at the end of this booklet; the question number or numbers must be clearly shown.
- You should use a calculator where appropriate.
- You must show all necessary working clearly; no marks will be given for unsupported answers from a calculator.
- Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place for angles in degrees, unless a different level of accuracy is specified in the question.

**INFORMATION**

- The total mark for this paper is 75.
- The number of marks for each question or part question is shown in brackets [ ].

This document has **20** pages. Any blank pages are indicated.

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- 2 (a) By sketching a suitable pair of graphs, show that the equation  $\cot 2x = \sec x$  has exactly one root in the interval  $0 < x < \frac{1}{2}\pi$ . [2]

- (b) Show that if a sequence of real values given by the iterative formula

$$x_{n+1} = \frac{1}{2} \tan^{-1}(\cos x_n)$$

converges, then it converges to the root in part (a). [1]

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- By first forming a quartic equation in  $x$  or  $y$ , find the square roots of  $6-8i$  in exact Cartesian form. [5]

[illegible]





5 (a) The complex number  $u$  is given by

$$u = \frac{(\cos \frac{1}{7}\pi + i \sin \frac{1}{7}\pi)^4}{\cos \frac{1}{7}\pi - i \sin \frac{1}{7}\pi}.$$

Find the exact value of  $\arg u$ .

[2]

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(b) The complex numbers  $u$  and  $u^*$  are plotted on an Argand diagram.

Describe the single geometrical transformation that maps  $u$  onto  $u^*$  and state the exact value of  $\arg u^*$ .

[2]

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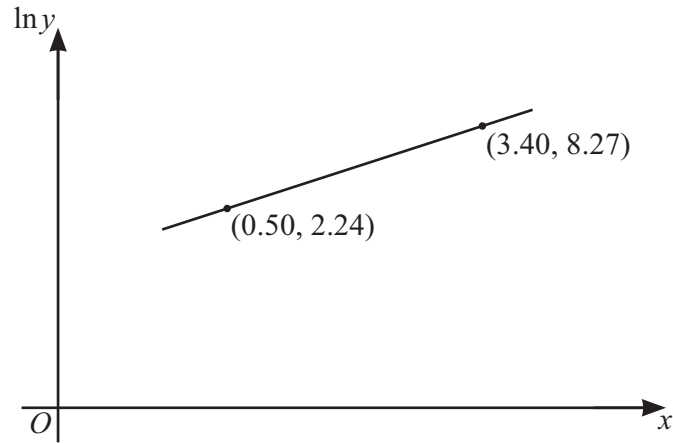
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The variables  $x$  and  $y$  satisfy the equation  $ay = b^x$ , where  $a$  and  $b$  are constants. The graph of  $\ln y$  against  $x$  is a straight line passing through the points  $(0.50, 2.24)$  and  $(3.40, 8.27)$ , as shown in the diagram.

Find the values of  $a$  and  $b$ . Give each value correct to 1 significant figure.

[4]

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- $$\tan^4 x - 2 \tan^2 x - 3 = 0$$

for  $\tan x \neq 0$ .

[3]

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- (b) Hence solve the equation  $\tan^3 2\theta + 2 \tan 4\theta - \tan 2\theta = 0$  for  $0 < \theta < \pi$ . Give your answers in exact form. [3]

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**8** The parametric equations of a curve are

$$x = \tan^2 2t, \quad y = \cos 2t,$$

for  $0 < t < \frac{1}{4}\pi$ .

(a) Show that  $\frac{dy}{dx} = -\frac{1}{2}\cos^3 2t$ .

[4]

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- (b)** Hence find the equation of the normal to the curve at the point where  $t = \frac{1}{8}\pi$ . Give your answer in the form  $y = mx + c$ . [4]

[illegible]



9 With respect to the origin  $O$ , the points  $A$ ,  $B$  and  $C$  have position vectors given by

$$\overrightarrow{OA} = \begin{pmatrix} 2 \\ 1 \\ -3 \end{pmatrix}, \quad \overrightarrow{OB} = \begin{pmatrix} 0 \\ 4 \\ 1 \end{pmatrix} \quad \text{and} \quad \overrightarrow{OC} = \begin{pmatrix} -3 \\ -2 \\ 2 \end{pmatrix}.$$

(a) The point  $D$  is such that  $ABCD$  is a trapezium with  $\overrightarrow{DC} = 3\overrightarrow{AB}$ .

Find the position vector of  $D$ .

[2]

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(b) The diagonals of the trapezium intersect at the point  $P$ .

Find the position vector of  $P$ .

[5]

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(c) Using a scalar product, calculate angle  $ABC$ . [4]

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- 10 A balloon in the shape of a sphere has volume  $V$  and radius  $r$ . Air is pumped into the balloon at a constant rate of  $40\pi$  starting when time  $t = 0$  and  $r = 0$ . At the same time, air begins to flow out of the balloon at a rate of  $0.8\pi r$ . The balloon remains a sphere at all times.

(a) Show that  $r$  and  $t$  satisfy the differential equation

$$\frac{dr}{dt} = \frac{50-r}{5r^2}. \quad [3]$$

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(b) Find the quotient and remainder when  $5r^2$  is divided by  $50 - r$ . [3]

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(c) Solve the differential equation in part (a), obtaining an expression for  $t$  in terms of  $r$ . [6]

[illegible]

**(d)** Find the value of  $t$  when the radius of the balloon is 12. [1]

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**11** Let  $f(x) = \frac{2e^{2x}}{e^{2x} - 3e^x + 2}$ .

- (a) Find  $f'(x)$  and hence find the exact coordinates of the stationary point of the curve with equation  $y = f(x)$ . [5]

[illegible]



- (b)** Use the substitution  $u = e^x$  and partial fractions to find the exact value of  $\int_{\ln 3}^{\ln 5} f(x) dx$ .

Give your answer in the form  $\ln a$ , where  $a$  is a rational number in its simplest form.

[9]

[illegible]

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